

FROM BASIC RESEARCH TO DISEASE MECHANISMS IN LIPID CELL BIOLOGY

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ABSTRACT

Eukaryotic cells produce over 1,000 different lipid species that tune organelle membrane properties, control signaling and store energy. Studying the biological functions of individual lipid has been challenging, owing to insufficient methodology for visualizing their intracellular localization, transport between organelles and interactions with proteins. Using minimally modified lipid probes, we set up an experimental platform for mechanistic lipid cell biology.

We used this approach to measure transport rates for individual lipids in cellular assays and discovered that non-vesicular lipid transport is the primary mechanism that the cell uses to establish the distinct lipid compositions of organelle membranes. Expanding this technology to enable detection of lipid-protein interactions, we mapped lipid-protein interactions during lipid trafficking, found that integral membrane protein components of both the MICOS and NPC complexes are positioned in highly curved membranes by interactions with phosphatidylethanolamine lipids and discovered the likely disease mechanism of ether lipid deficiencies. We anticipate that chemical biology approaches will become an integral component of lipid biology and catalyze further discoveries.